STI WEST NEGROS UNIVERSITY

College of Information and Communications Technology

SmartTrack: A Mobile Attendance Monitoring System Using Biometric & Dynamic QR Verification

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Chapters 1 to 3: Problem Background, Literature Review, and Technical System Methodology

BSIT Capstone Project Manuscript

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Chapter 1: The Problem and Its Background

Manual roll-call and paper sign-in sheets remain pervasive across university classrooms despite widespread digital transformation. These legacy methods consume up to 15 minutes of scheduled lecture time, are vulnerable to fraudulent proxy attendance, and require labor-intensive manual data re-entry into institutional learning management systems.

1.1 Project Objectives

The primary objective is to architect and deploy SmartTrack, a cross-platform mobile application combining dynamic time-sensitive QR codes with biometric fingerprint/face authentication. Specific aims include: 1) eliminating paper waste and human recording errors; 2) synchronizing real-time classroom records with university administrative databases; and 3) alerting academic advisors when students breach absentee warning thresholds.

1.2 Scope and Delimitations

The system scope covers enrolled students and instructors within the College of Information and Communications Technology at STI West Negros University. Delimitations restrict offline attendance logging to localized caching, requiring an active internet handshake within 12 hours for official institutional ledger commitment.

TECHNICAL INNOVATION HIGHLIGHT: Dynamic QR codes refresh every 10 seconds with cryptographic hashing, completely preventing unauthorized off-campus photo sharing and proxy check-ins.

Chapter 2: Review of Related Literature and Studies

Previous academic literature has explored RFID badges, Bluetooth Low Energy (BLE) beacons, and facial recognition terminals. Patel and Priyadarshi (2020) demonstrated that while fixed biometric readers provide reliable verification, physical bottleneck queues at classroom doorways severely disrupt bell schedules.

Static QR code systems, while inexpensive, suffer from severe proxy fraud vulnerabilities where students photograph code slips and broadcast them over social messaging channels to absent peers.

Table 1. Comparative Analysis of Academic Attendance Technologies

Technology Framework	Hardware Cost	Queue Bottleneck	Proxy Vulnerability
Paper Sign-in Logbooks	Zero Initial Cost	Severe Delay	Extremely High (Proxy Signing)

Wall-Mounted Fingerprint	High (P15,000/room)	High Physical Queue	Zero (Biometric Tied)
Static Printed QR Code	Minimal (Paper QR)	Low Delay	High (Image Shared on Chat)
SmartTrack Dynamic QR	Low (Existing Phones)	Zero Bottleneck (<3 sec)	Zero (Rotating Hash + Biometric)

2.1 Conceptual Framework: Input-Process-Output

Under the IPO model, Inputs comprise student credentials, geo-fenced campus coordinates, and classroom session tokens. The Process stage encompasses cryptographic SHA-256 rotating QR rendering, biometric enclave authentication, and timestamp verification. The Output is an authenticated attendance record reflected instantly on instructor dashboards.

Chapter 3: Research Methodology and Technical Design

The project adopts the Agile Scrum methodology, consisting of four two-week sprints: Requirements Gathering, UI/UX Wireframing in Figma, REST API and Database Architecture, and System Testing. The backend is built upon Node.js and PostgreSQL, with client applications built in Flutter for simultaneous Android and iOS support.

Data privacy is maintained in strict compliance with Republic Act No. 10173 (Data Privacy Act of 2012). Biometric templates remain stored within the client smartphone's secure enclave and are never transmitted across university network servers.

3.1 System Testing and Acceptance Protocols

Evaluation involves three rigorous testing tiers: Unit Testing with Jest for cryptographic token validity, Load Testing with k6 simulating 500 simultaneous requests per minute during class changeovers, and User Acceptance Testing (UAT) evaluated using ISO 9126 software quality metrics.

References

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